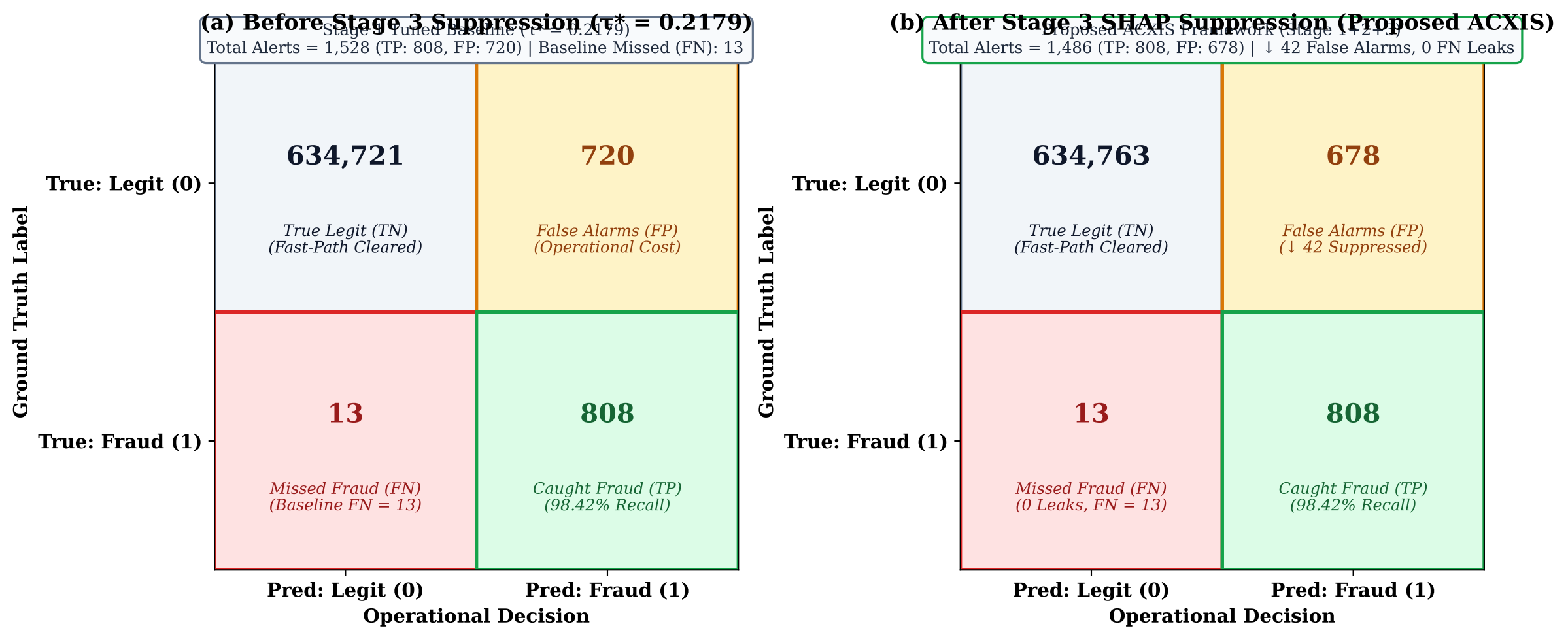
(a) Before Stage 3 Suppression (τ* = 0.2179)
Stage 3 Tuned Baseline (τ* = 0.2179)
Total Alerts = 1,528 (TP: 808, FP: 720) | Baseline Missed (FN): 13
634,721
True Legit (TN)
(Fast-Path Cleared)
720
False Alarms (FP)
(Operational Cost)
13
Missed Fraud (FN)
(Baseline FN = 13)
808
Caught Fraud (TP)
(98.42% Recall)
True: Legit (0)
True: Fraud (1)
Ground Truth Label
Pred: Legit (0)
Pred: Fraud (1)
Operational Decision
(b) After Stage 3 SHAP Suppression (Proposed ACXIS)
Proposed ACXIS Framework (Stage 1+2+3)
Total Alerts = 1,486 (TP: 808, FP: 678) | ↓ 42 False Alarms, 0 FN Leaks
634,763
True Legit (TN)
(Fast-Path Cleared)
678
False Alarms (FP)
(↓ 42 Suppressed)
13
Missed Fraud (FN)
(0 Leaks, FN = 13)
808
Caught Fraud (TP)
(98.42% Recall)
True: Legit (0)
True: Fraud (1)
Ground Truth Label
Pred: Legit (0)
Pred: Fraud (1)
Operational Decision